SCHOOL OF COMPUTER SCIENCE

LEVEL 3 COMP3038 AUTUMN SEMESTER 2025-2026

Machine Learning Module Exam

Instructions:

Time allowed: TWO Hours

Answer All 3 Questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

Only a calculator from approved list A may be used in this examination.

List A

Basic Models	***Scientific Calculators***
Aurora HC133 Casio HS-5D Deli – DL1654 Sharp EL-233	Aurora AX-582 Casio FX82 family Casio FX83 family Casio FX85 family Casio FX350 family Casio FX570 family Casio FX 991 family Sharp EL-531 family Texas Instruments TI-30 family Texas BA II+ family

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: None

Additional material: None

Information for Invigilators: None
Exam Papers must not be removed from the examination venue

Question 1 : Foundations of Machine Learning [overall 18 marks]

There are many different types of credit card fraud, but at its simplest, it's when someone obtains your card details and makes transactions on your card without you knowing.

a) Explain what the credit card fraud transaction problem is in the context of machine learning for this simplest scenario and what T, P & E are.

[7 marks]

b) Give the formal definition of overfitting. Explain when the overfitting occurs and why overfitting is such a big problem for Machine Learning?

[5 marks]

c) We trained two classifiers, Logistic Regression with 300 examples and tested with 30 examples; and the Support Vector Machine with 400 examples and tested with 50 examples. The correctly classified percentage of the Logistic Regression is 82% and for the support vector machine it is 92%. Calculate the confidence intervals of the two true errors and comment how that relates to the statistical difference between the results. Z_n= 1.96 for a confidence level of 95%.

[6 marks]

Name: **Student ID:**

Answers: